



Agentic Skill Lifecycle

From reusable prompt to governed platform capability

20 minutes

Platform Engineering

Skills + Agentic Workflows



How do we make agent capabilities reusable across every execution surface without losing control?



Developer

Local client



Cloud agent

Repository context



Workflow runner

Ephemeral runtime



Reviewer

Evidence + approval

A capability working on one laptop is not yet a platform capability.

1 What kind of thing is it?

Instruction

Repository truth

Skill

Reusable procedure

Agent

Role + tools

Workflow

Event + controls

MCP / tool

External capability

Ruleset

Enforcement



2 Who needs it?

User

personal lifetime

Repository

project lifetime

Workflow run

ephemeral lifetime

Org / enterprise

where supported

Context is not enforcement. Instructions and skills guide behavior; permissions, tests, rulesets, and humans enforce boundaries.



CONTEXT

What the agent knows

Repository instructions

Relevant skills

Agent role + tools



EXECUTION

Where reasoning runs

Local client

Cloud agent

Agentic Workflow

↓ **proposed output**



GOVERNANCE

What may change

Tests

Permissions

Safe outputs

Rulesets

Human review



1. Author

Canonical source



2. Validate

Structure + tests



3. Release

Immutable identity



4. Approve

Preview + review



5. Deliver

Minimum lane



6. Execute

Bounded runtime



7. Observe

Evidence + drift



8. Evolve

Upgrade · rollback · retire

Every transition leaves evidence

source review

validation result

tag / SHA

provenance

consumer PR

runtime output

Release is a checkpoint — **not the finish line.**

REFERENCE REPOSITORY

skills/

└ generate-design/

| └ SKILL.md

| └ references/

└ implement-feature/

| └ SKILL.md

└ review-code/

| └ SKILL.md

SKILL.md is the portable procedure. References, scripts, and assets load only when needed.

Discovery contract

The description tells clients **when** to load the skill. Vague descriptions create missed or incorrect activation.



Structure

Name, metadata, paths



Neutrality

Proof-specific portability check



Unit tests

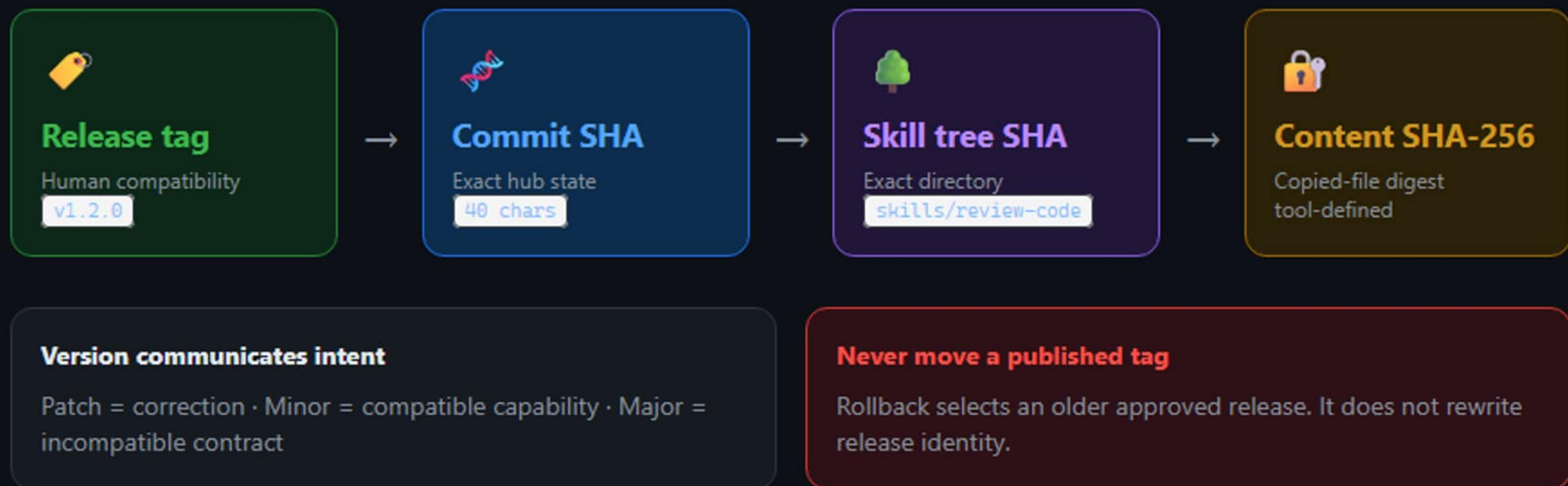
Deterministic helpers



Publish dry run

Release readiness

Source of truth: change the hub → release a version → promote the artifact. Do not maintain consumer forks by hand.





persistent

User-installed

- ✓ One developer + client
- ✓ Personal productivity
- ✓ Early evaluation
- ✗ Not visible to remote runners



ephemeral

Workflow runtime

- ✓ One compiled workflow run
- ✓ Full commit-SHA pin
- ✓ No repository vendoring
- ✗ Not repository context



versioned

Repository-installed

- ✓ Team + cloud agents
- ✓ Reviewed with code
- ✓ Repository provenance
- ✗ Must manage drift

Choose the **minimum lane required by the consumer.**

More copies create more upgrade work and more opportunities for drift.



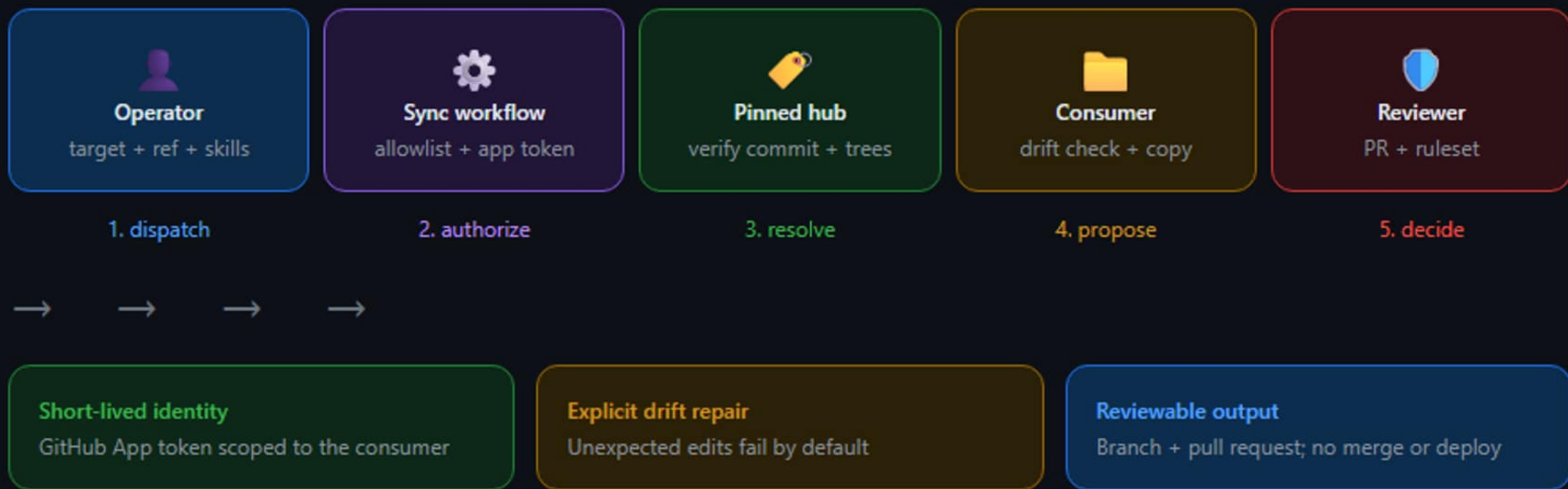
Release

```
gh skill publish --dry-run  
gh skill publish --tag v1.1.0
```

Trust + install

```
gh skill preview OWNER/HUB review-code@v1.1.0  
gh skill install OWNER/HUB review-code --pin v1.1.0
```

gh skill preview feature: verify command behavior, supported clients, and destination paths with your installed GitHub CLI before production rollout.



`.agent-skills-lock.json` is proof-specific platform inventory — not a built-in `gh skill` lock file.

HUMAN-READABLE SOURCE



workflow.md



gh aw compile

GENERATED IMPLEMENTATION



workflow.lock.yml

⚙️ Deterministic rail

Checkout

Install

Build

Test

Validate

Artifacts

🧠 Agentic rail

Investigate

Classify

Design

Explain

Bounded edit

Summarize

read-only reasoning

minimum permissions

network allowlist

declared safe outputs

A skill supplies procedure context. It does not grant authority.



Moving branch or tag

Same label, different behavior



Immutable release

Protect tags; publish or select another version



Editing installed copies

Drift and unclear ownership



Fail visibly

Fix the hub or explicitly repair drift



Local-only proof

False confidence across surfaces



Surface tests

Same approved identity in local, cloud, and Actions



Broad write token

Large prompt-injection blast radius



Bounded authority

Read-only agent + safe outputs + app token



CODEOWNERS alone

Reviewer requested, merge still bypassable



Enforcing ruleset

Require code-owner approval before merge

Drift is information. Do not silently erase it.



Personal productivity

User-installed pinned skill



Team + cloud consistency

Repository-installed skill



Workflow-only automation

Runtime-pinned skill



Platform-scale rollout

Hub + synchronization PRs

START SMALL

1

One repeated procedure

2

One canonical source

3

One immutable release

4

One required delivery lane

5

Evidence + rollback path

Source it. Test it. Version it. Distribute it.
Observe it. Govern it.

Reference implementation: github.com/cajetzer/agent-primitives-hub



OFFICIAL COMMAND REFERENCE

[cli.github.com/manual/gh skill](https://cli.github.com/manual/gh_skill)

`gh skill preview`

`gh skill install`

`gh skill publish`

`gh skill update`

The `gh skill` command family is in preview and subject to change.